Abstract Linear Algebra

Prof. Lanier
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Pick up a handout. Try those warm-up problems.

Next up:

- HW7 due tonight.
- Open office hours today, 3:30-4:30pm in Eckhart 207A
- Small group office hours continue through this week
- Last requizzing session this Friday, 3-5pm in Ryerson 177.
  (or maybe a bigger room)
- Flex (or flex plan) due by end-of-day Friday
- Final Exam review sessions: TBA (packet)
  TBA (2021 final)
- Final Exam: Both exams will be held in our classroom, Ryerson 276.
  Section 45 (11:30am section) - Thursday, May 25, 5:30-7:30pm
  Section 55 (12:30pm section) - Wednesday, May 24, 7:30-9:30am

Ch. 5 & 6: “Now with orthogonality!”

Row reduction:
change basis to make
an upper triangular matrix

LU decomposition:
A=LU

diagonalization:
$A = PDP^{-1}$
when A has n lin ind eigenvectors

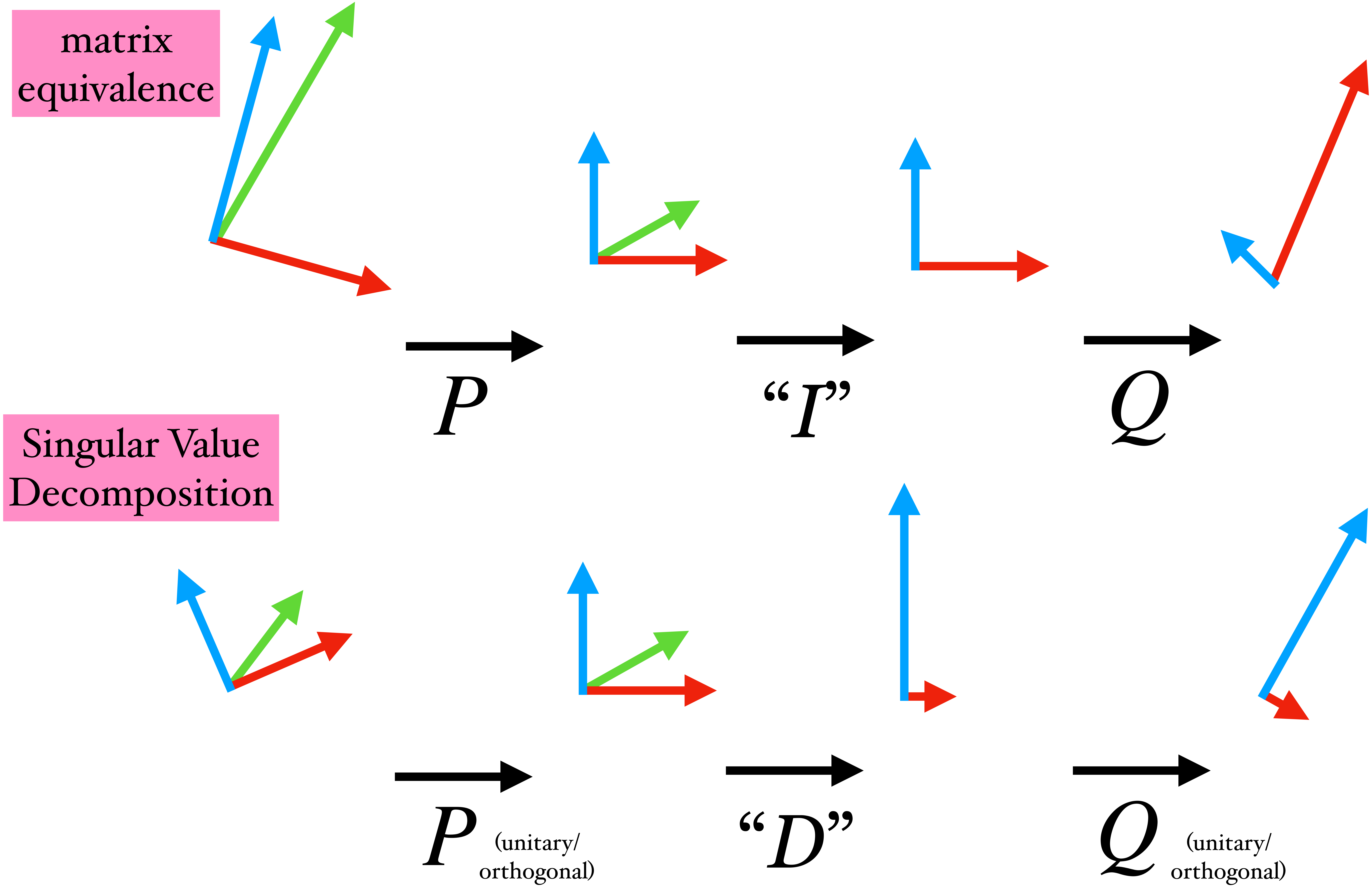
matrix equivalence:
A= Q“I” P, with Q & P invertible

Gram–Schmidt:
change basis to make
an orthogonal matrix

QR decomposition:
A=QR

Spectral Theorem:
$A = PDP^{-1}$ with real entries for D and
with P unitary (orthogonal, for $\mathbb{R}$)
when A is self-adjoint (symmetric for $\mathbb{R}$)
$A = A^*$

Singular Value Decomposition
A= Q“D”P, with Q & P unitary (orthogonal)


matrix equivalence
P
"I"
Q
Singular Value Decomposition
P (unitary/ orthogonal)
"D"
Q (unitary/ orthogonal)

Singular Value Decomposition
A= Q“D”P, with Q & P orthogonal
(or for $\mathbb{C}$, unitary)

Preliminary:

A^*A is self-adjoint (symmetric).

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By Spectral Theorem, A^*A is orthogonally diagonalizable.
Let λ_i and v_i be the (real, ordered) eigenvalues and orthonormal eigenbasis for $A^*A = PDP^{-1}$.

Singular Value Decomposition
A= Q"D"P, with Q & P orthogonal
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$$A^*A = \underbrace{\begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}}_{P} \underbrace{\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}^{-1}}_{P^{-1}}$$

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$k \leq n$
so that image vectors
are linearly independent

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And not every square matrix can be diagonalized.

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Every matrix has an SVD.
Every matrix A ($m \times n$) has singular values, $\sigma_1, \ldots, \sigma_n$.
At most m of them are nonzero.
They are equal to the square roots of the eigenvalues of the matrix A^*A .
The corresponding pairs of vectors in the domain and range (columns of P & Q)
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The SVD of a matrix is unique, up to orthogonal symmetries in the domain and codomain.
(permuting columns and rotations/reflections within singular spaces)
In particular, the singular values are a well-defined invariant of a matrix.

A square matrix has eigenvalues and singular values. How are they related?

λ_i σ_i

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Of course, they can be: the eigenvalues and singular values of any diagonal matrix are equal,
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Weyl inequality: $|\lambda_1(A) \ldots \lambda_k(A)| \le \sigma_1(A) \ldots \sigma_k(A)$ for $k = 1, \ldots, n$
and equality for $k = n$.

Example comparing eigenvalues and singular values:

Diagonalization $\qquad\qquad A \qquad\qquad$ SVD

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 6 & 0 \\ 0 & 7 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 6 & 0 \\ 0 & 7 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Example comparing eigenvalues and singular values:

Diagonalization $\qquad\qquad A \qquad\qquad$ SVD

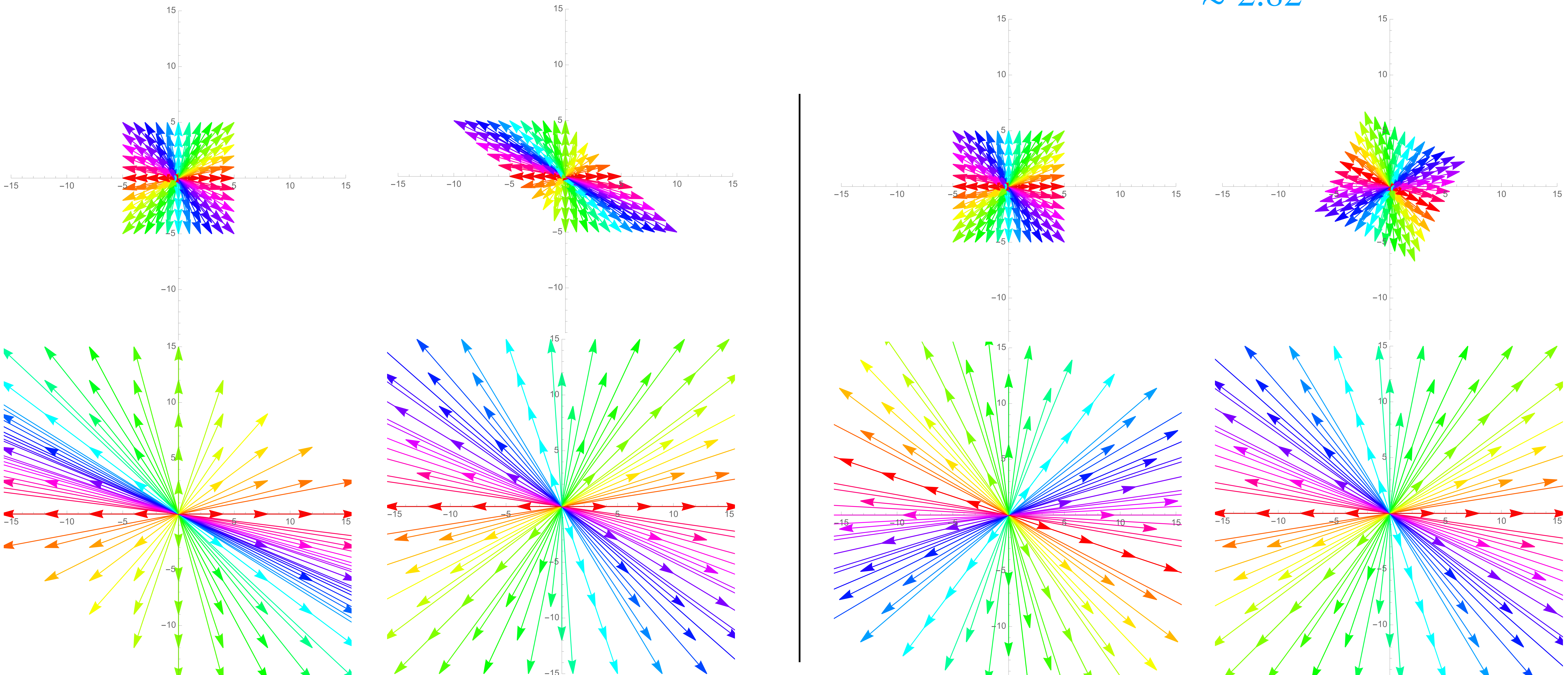
$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 6 & 0 \\ 0 & 7 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 6 & 0 \\ 0 & 7 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} -\frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{pmatrix}\begin{pmatrix} \overset{\approx 4.24}{3\sqrt{2}} & 0 \\ 0 & \underset{\approx 2.82}{2\sqrt{2}} \end{pmatrix}\begin{pmatrix} -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}$$

Diagonalization

SVD

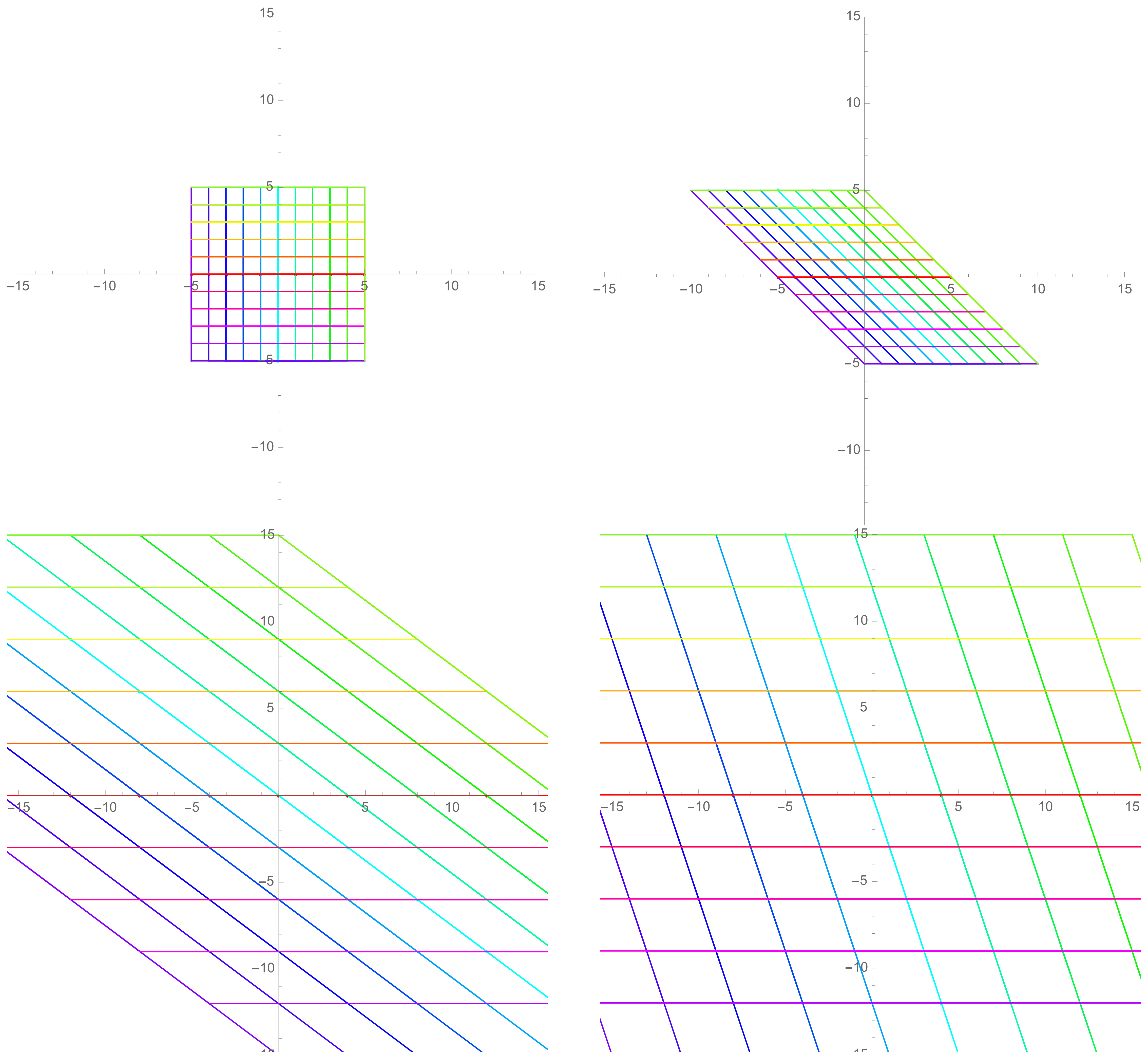
$$\begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}\begin{pmatrix}4 & 0\\0 & 3\end{pmatrix}\begin{pmatrix}1 & -1\\0 & 1\end{pmatrix}=\begin{pmatrix}4 & 1\\0 & 3\end{pmatrix}=\begin{pmatrix}-\frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}}\\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}}\end{pmatrix}\begin{pmatrix}\overset{\approx 4.24}{3\sqrt{2}} & 0\\0 & \underset{\approx 2.82}{2\sqrt{2}}\end{pmatrix}\begin{pmatrix}-\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}}\\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}}\end{pmatrix}$$

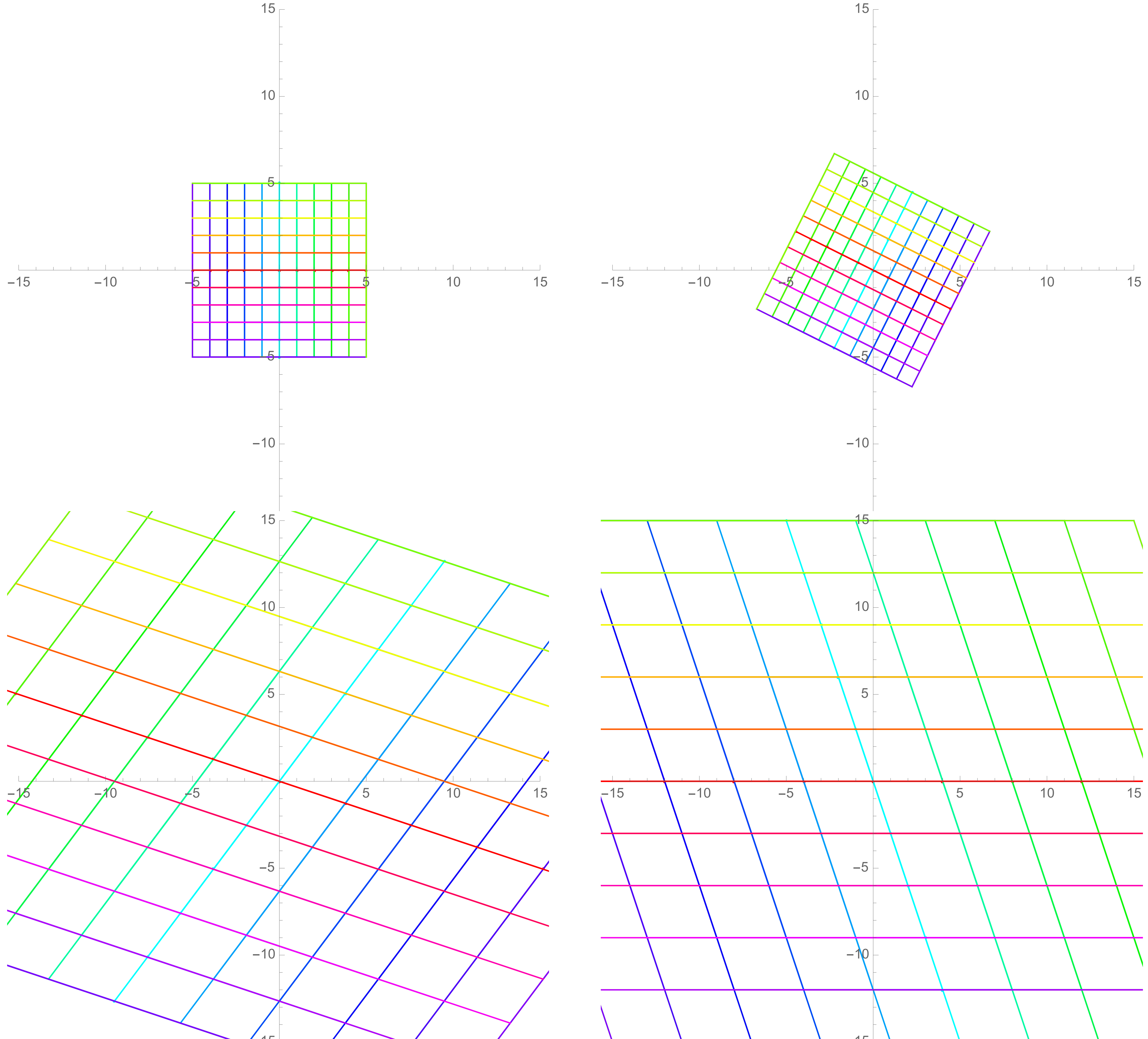


Diagonalization

SVD

$$\begin{pmatrix}1&1\\0&1\end{pmatrix}\begin{pmatrix}4&0\\0&3\end{pmatrix}\begin{pmatrix}1&-1\\0&1\end{pmatrix}=\begin{pmatrix}4&1\\0&3\end{pmatrix}=\begin{pmatrix}-\frac{3}{\sqrt{10}}&\frac{1}{\sqrt{10}}\\ \frac{1}{\sqrt{10}}&\frac{3}{\sqrt{10}}\end{pmatrix}\begin{pmatrix}\overset{\approx 4.24}{3\sqrt{2}}&0\\0&\underset{\approx 2.82}{2\sqrt{2}}\end{pmatrix}\begin{pmatrix}-\frac{2}{\sqrt{5}}&\frac{1}{\sqrt{5}}\\ \frac{1}{\sqrt{5}}&\frac{2}{\sqrt{5}}\end{pmatrix}$$





An arithmetic-geometric inequality for λ_i and σ_i
(Raised by Bhatia–Kittaneh in 1990, proved by Sam Drury in 2012)

$$\sqrt{ab} \leq \frac{a+b}{2} \qquad \text{for non-negative } a, b \in \mathbb{R}$$

An arithmetic-geometric inequality for λ_i and σ_i
(Raised by Bhatia–Kittaneh in 1990, proved by Sam Drury in 2012)

$$\sqrt{ab} \le \frac{a+b}{2}$$ for non-negative $a, b \in \mathbb{R}$

$$\sqrt{\sigma_i(PQ)} \le \frac{\lambda_i(P+Q)}{2}$$ for positive semidefinite $n \times n$ matrices P, Q

(all eigenvalues non-negative)

$$\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}^2 - 5\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix} + 6\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}^2 - 5\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix} + 6\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 10 \\ -5 & 14 \end{pmatrix} - \begin{pmatrix} 5 & 10 \\ -5 & 20 \end{pmatrix} + \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$$

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$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{vmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{vmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}^2 - 5\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix} + 6\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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$$\begin{vmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda) + 2 = \lambda^2 - 5\lambda + 6$$

$$\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}^{2} - 5\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix} + 6\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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$$\begin{vmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda) + 2 = \lambda^2 - 5\lambda + 6$$

Is every square matrix a “zero” of its characteristic polynomial?

diagonal matrices?

$$\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}$$

diagonal matrices?

$$\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} \qquad (5-\lambda)(4-\lambda) \qquad = \lambda^2 - 9\lambda + 20$$

diagonal matrices?

$$\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} \qquad (5-\lambda)(4-\lambda) \qquad = \lambda^2 - 9\lambda + 20$$

$$(A - 5I)(A - 4I)$$

diagonal matrices?

$$\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} \qquad (5-\lambda)(4-\lambda) \qquad = \lambda^2 - 9\lambda + 20$$

$$(A - 5I)(A - 4I)$$

$$\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

diagonal matrices?

$$\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} \qquad (5-\lambda)(4-\lambda) \qquad = \lambda^2 - 9\lambda + 20$$

$$(A - 5I)(A - 4I)$$

$$\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$$

diagonal matrices?

$$\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}$$

$$(5 - \lambda)(4 - \lambda) \qquad = \lambda^2 - 9\lambda + 20$$

$$(A - 5I)(A - 4I)$$

$$\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \quad = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a \\ 0 \end{pmatrix}$$

diagonal matrices?

$$\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} \qquad (5-\lambda)(4-\lambda) \qquad = \lambda^2 - 9\lambda + 20$$

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$$\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \quad = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a \\ 0 \end{pmatrix} \quad = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

diagonal matrices?

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$$(A-5I)(A-4I)$$

$$\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} a \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Next up:

- HW7 due tonight.
- Open office hours today, 3:30-4:30pm in Eckhart 207A
- Small group office hours continue through this week
- Last requizzing session this Friday, 3-5pm in Ryerson 177.
  (or maybe a bigger room)
- Flex (or flex plan) due by end-of-day Friday
- Final Exam review sessions: TBA (packet)
  TBA (2021 final)
- Final Exam: Both exams will be held in our classroom, Ryerson 276.
  Section 45 (11:30am section) - Thursday, May 25, 5:30-7:30pm
  Section 55 (12:30pm section) - Wednesday, May 24, 7:30-9:30am